

Problem

The current through a 25-mH inductor is $10e^{-t/2}$ A. Find the voltage and the power at $t = 3$ s.

Step-by-step solution

Step 1 of 2

The current through the inductor is, $i = 10e^{-\frac{t}{2}}$ A

Determine the voltage across the inductor.

Substitute 25 mH for L and $10e^{-\frac{t}{2}}$ A for i .

$$\begin{aligned} v_L &= L \frac{di}{dt} \\ &= (25 \times 10^{-3}) \frac{d}{dt} \left(10e^{-\frac{t}{2}} \right) \\ &= (25 \times 10^{-3}) \left(-\frac{1}{2} \right) \left(10e^{-\frac{t}{2}} \right) \\ &= -125e^{-\frac{t}{2}} \text{ mV} \end{aligned}$$

Substitute 3 s for t in the equation.

$$\begin{aligned} v_L &= -125e^{-\frac{3}{2}} \text{ mV} \\ &= -27.89 \text{ mV} \end{aligned}$$

Therefore, the voltage across the inductor at $t = 3$ s is $\boxed{-27.89 \text{ mV}}$.

Step 2 of 2

Determine the value of power at $t = 3$ s.

$$\begin{aligned} p &= v_L i \\ &= \left(-125e^{-\frac{3}{2}} \text{ mV} \right) \left(10e^{-\frac{t}{2}} \right) \end{aligned}$$

Substitute 3 s for t in the equation.

$$\begin{aligned} p &= \left(-125e^{-\frac{3}{2}} \text{ mV} \right) \left(10e^{-\frac{3}{2}} \right) \\ &= (-27.89)(2.231) \\ &= -62.22 \text{ mW} \end{aligned}$$

Therefore, the value of power at $t = 3$ s is $\boxed{-62.22 \text{ mW}}$.